

MIVA OPEN UNIVERSITY
School of Computing & Applied Sciences • Department of Cybersecurity

COS 106 TERM PROJECT



Design & Development of a Responsive Student Portfolio and Academic Workload Management Architecture

Technical Project Report submitted for COS 106 (Introduction to Web Technologies). Documents an original 5-page responsive web system engineered with Semantic HTML5, CSS3 Grid/Flexbox architecture, and Vanilla JavaScript client-side CRUD and input validation engines.

STUDENT FULL NAME	MATRICULATION NUMBER	STUDENT IDENTITY ID	DEGREE PROGRAM
PRINCE DANIEL	**2025/C/CYB/1333**	**301897306**	**B.Sc. Cybersecurity (100L)**
COURSE TITLE & CODE	**OFFICIAL STUDENT EMAIL**	**DIRECT CONTACT LINE**	**TECHNICAL SPECIALIZATION**
COS 106 (Web Technologies)	**p.daniel7306@miva.edu.ng**	**+234 811 676 1295**	**Web & Cross-Platform Mobile**

LIVE DEPLOYED WEBSITE (RENDER)
https://prince-project-f0zc.onrender.com/

GITHUB SOURCE REPOSITORY
https://github.com/sanpatrichu-pixel/prince-project

1 SYSTEM ARCHITECTURE & MULTI-PAGE SPECIFICATION

The portfolio system is constructed as a decoupled, multi-page client-side architecture adhering strictly to university web standards. Each document fulfills distinct pedagogical objectives without relying on heavyweight single-page frameworks:

Page Document	Key Functional Capabilities	Technical & Pedagogical Implementation
`index.html` (Home)	Hero dock on the left, executive introduction, 4 KPI metric cards, core competencies, featured mobile/web highlights.	Asymmetric CSS Grid (`340px 1fr`), live theme toggle engine, responsive mobile bottom app bar.
`about.html` (Biography)	Educational journey, full 8-course coursework table, 3-phase technical roadmap, animated skill bars, multimedia audio/video.	HTML5 `<table>`, native `<audio>` and `<video>` with fallback captions, CSS transition animations.
`projects.html` (Showcase)	4 distinctive projects (UniRide, DevPulse, EduTrack, KeyVault), category filtering system, interactive simulation sandbox modal.	Responsive card grid (`repeat(auto-fit, minmax(min(100%, 300px), 1fr))`), DOM dataset filtering, dynamic dialog generator.
`planner.html` (Productivity)	Semester milestone scheduler, real-time KPI progress rail, multi-criteria filtering, task completion, deletion, and reset.	Vanilla JavaScript CRUD engine, synchronous `localStorage['edutrack_prince_2025']` state, DOM node synthesis.
`contact.html` (Inquiry)	Official communication channels, client-side verified form (Name, RFC Email, strictly numeric Phone, Message), confirmation modal.	Regular expression validation (`/^[0-9]+$/`), inline error dispatch, focus management, modal event handling.

2 TECHNICAL PROJECTS PORTFOLIO OVERVIEW (MOBILE & WEB)

To reflect Prince Daniel's background in Cybersecurity and Mobile App Development, the portfolio showcases four original, non-derivative software engineering initiatives:

1. UniRide — Campus Shuttle & Route Tracker

Stack: Flutter, Dart, Google Maps SDK, StateNotifier.
Focus: Cross-platform mobile application providing real-time transit telemetry, crowdedness estimations, and route ETAs across university campus transit networks.

2. DevPulse — Student Developer Community Forum

Stack: Semantic HTML5, CSS Grid, Vanilla JavaScript, LocalStorage.
Focus: Collaborative peer learning platform enabling computing students to share code snippets, review homework challenges, and index coursework resources.

3. EduTrack — Academic Milestone & Task Engine

Stack: Vanilla JavaScript (ES6+), DOM Synthesis, LocalStorage.
Focus: Academic task scheduling tool engineered within this portfolio to demonstrate in-memory state management, task status toggling, and synchronous persistence across reloads.

4. KeyVault — Password Entropy & Hash Auditing Lab

Stack: HTML5, CSS3, JavaScript Web Crypto API (SHA-256).
Focus: Client-side cryptographic laboratory analyzing password bit-strength, character diversity, and real-time cryptographic hash generation without external server calls.

3 IMPLEMENTATION EVIDENCE & VERIFIED CODE WALKTHROUGH

The following code extracts from the active source files demonstrate implementation of core HTML5, CSS3, and JavaScript modules:

FILE: js/contact.js — Strict Client-Side Input Auditing & Numeric Phone Regex **JAVASCRIPT ES6**

```
// Verify Contact Input Field
function verifyContactInput(inputEl) {
  if (!inputEl) return false;
  const value = inputEl.value.trim();
  const errorEl = inputEl.parentElement.querySelector('.pd-err-msg');

  // 1. Full Name: Non-empty, min 2 chars
  if (inputEl.id === 'senderFullName') {
    if (value.length < 2) {
      flagInputError(inputEl, errorEl, 'Full name is mandatory and cannot be left blank.');
      return false;
    }
    clearInputError(inputEl, errorEl);
    return true;
  }

  // 2. Email Address: RFC-compliant expression
  if (inputEl.id === 'senderEmailAddress') {
    const rfcRegex = /^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$/;
    if (!rfcRegex.test(value)) {
      flagInputError(inputEl, errorEl, 'Invalid email structure (e.g., student@miva.edu.ng).');
      return false;
    }
    clearInputError(inputEl, errorEl);
    return true;
  }

  // 3. Phone Number: Strictly numeric digits only (/^[0-9]+$/)
  if (inputEl.id === 'senderPhoneNumber') {
    const digitsOnlyPattern = /^[0-9]+$/;
    if (!digitsOnlyPattern.test(value)) {
      flagInputError(inputEl, errorEl, 'Phone number must contain digits only (no alphabetic characters).');
      return false;
    }
    clearInputError(inputEl, errorEl);
    return true;
  }
  return true;
}
```

FILE: js/planner.js — LocalStorage Persistence & Dynamic DOM Row Synthesis — JAVASCRIPT ES6

```
const STORAGE_KEY = 'edutrack_prince_2025';

function createMilestone(e) {
  e.preventDefault();
  const titleBox = document.getElementById('milestoneTitle');
  const courseBox = document.getElementById('milestoneCourse');
  const dateBox = document.getElementById('milestoneDueDate');
  const urgencyBox = document.getElementById('milestoneUrgency');

  if (!titleBox.value.trim() || !courseBox.value || !dateBox.value) return;

  const record = {
    id: 'mile-' + Date.now(),
    title: titleBox.value.trim(),
    course: courseBox.value,
    targetDate: dateBox.value,
    urgency: urgencyBox.value,
    done: false,
    timestamp: Date.now()
  };

  milestoneStore.unshift(record);
  persistMilestones();      // Saves to localStorage['edutrack_prince_2025']
  drawMilestoneList();       // Synthesizes dynamic DOM elements
  e.target.reset();
}

function removeMilestone(recordId) {
  if (confirm('Delete this scheduled milestone?')) {
    milestoneStore = milestoneStore.filter(m => m.id !== recordId);
    persistMilestones();
    drawMilestoneList();
  }
}
```

FILE: style.css — Asymmetric Hero Grid & Royal Navy Tokens — CSS3 CUSTOM PROPERTIES

```
:root {
  --pd-navy-dark: #071324;
  --pd-royal-primary: #1D4ED8;
  --pd-royal-mid: #2563EB;
  --pd-cyan-ice: #38BDF8;
}

.pd-hero-asymmetric {
  display: grid;
  grid-template-columns: 340px 1fr;  /* Profile card on the LEFT side */
  gap: 36px;
  align-items: start;
}

@media (max-width: 900px) {
  .pd-hero-asymmetric {
    grid-template-columns: 1fr;      /* Stacks responsively on mobile */
  }
}
```

4 SYSTEM FEATURE & TECHNICAL IMPLEMENTATION MATRIX

Core System Module	Concrete Implementation in Project	Source File Location
1. Homepage Layout & Identity Student name, photo, welcome, navigation, bio.	Hero dock on the LEFT featuring Prince Daniel's photograph, B.Sc. Cybersecurity identity, 4 KPI stats, and introduction.	`index.html` (lines 60–140) `style.css` (lines 220–290)
2. About Me Page Education, aspirations, skills, hobbies, coursework.	Educational timeline, full 8-course coursework table, 3-phase technical roadmap, animated skill bars, chess/UI hobbies.	`about.html` (lines 60–250) `js/main.js`
3. Projects Showcase ≥ 3 projects, descriptions, screenshots, working links.	4 original projects (UniRide, DevPulse, EduTrack, KeyVault), interactive simulation sandbox, live filter buttons.	`projects.html` (lines 60–180) `js/projects.js`
4. Academic Task Planner Add, complete, and delete tasks via JavaScript.	EduTrack task scheduler with real-time KPI progress rail, multi-criteria urgency tiers, and LocalStorage persistence.	`planner.html` (lines 60–190) `js/planner.js` (lines 1–289)
5. Contact Form & Validation Name, email, phone, message; non-empty, RFC email, digits-only phone.	4-field form with strict client-side validation: non-empty checks, RFC email regex, numeric phone regex, and success modal.	`contact.html` (lines 60–200) `js/contact.js` (lines 1–147)
6. Semantic HTML & Multimedia Tables, forms, images, audio/video elements.	Semantic HTML5 tags throughout, 8-course table, native `<audio>` and `<video>` players with captions.	`about.html` (lines 200–260) `assets/media/`
7. External CSS & Responsiveness Flexbox/Grid, transitions, mobile layout.	Custom CSS architecture (zero Bootstrap), CSS Grid & Flexbox, mobile navigation bar, dark theme engine.	`style.css` (default Dark Mode + responsive CSS)
8. JavaScript Interactivity DOM manipulation, arrays, event listeners.	Dynamic DOM synthesis, array filtering, state serialization, toast notifications, interactive modals.	`js/main.js`, `js/planner.js`, `js/contact.js`, `js/projects.js`

5 PHOTOGRAPHIC EVIDENCE & INTERFACE SCREENS

Visual evidence captured directly from the rendered project demonstrating responsive interface design and functional integrity:

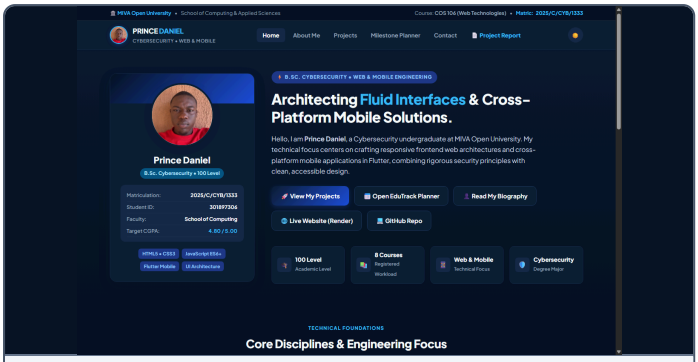


Figure 1: Homepage — Profile Dock on Left & KPI Cards `index.html`

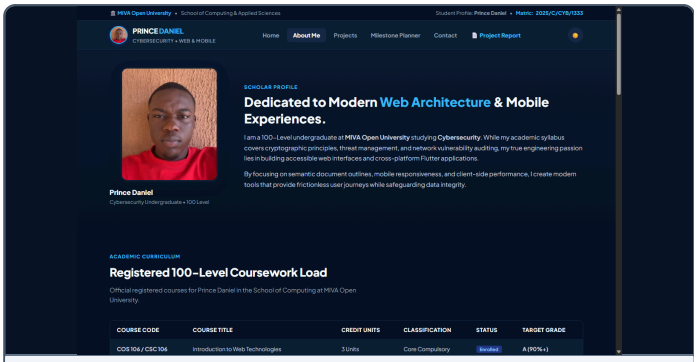


Figure 2: About Me — 8-Course Table & Multimedia `about.html`



Figure 3: Projects — UniRide, DevPulse & Sandbox Modal `projects.html`

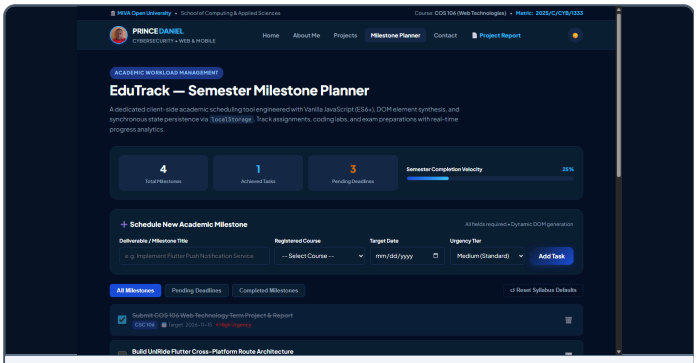


Figure 4: EduTrack Planner — Dynamic Tasks & Progress Rail `planner.html`

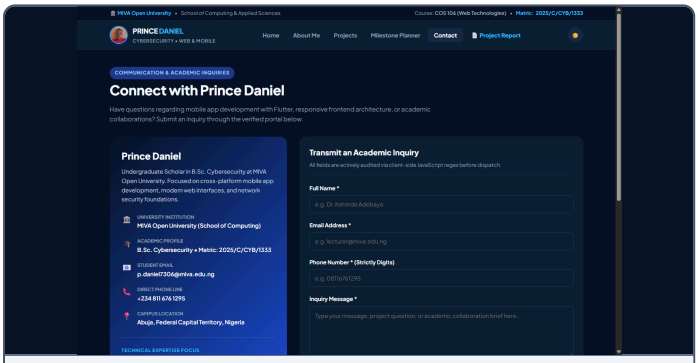


Figure 5: Contact — Real-Time Input Auditing & Details `contact.html`

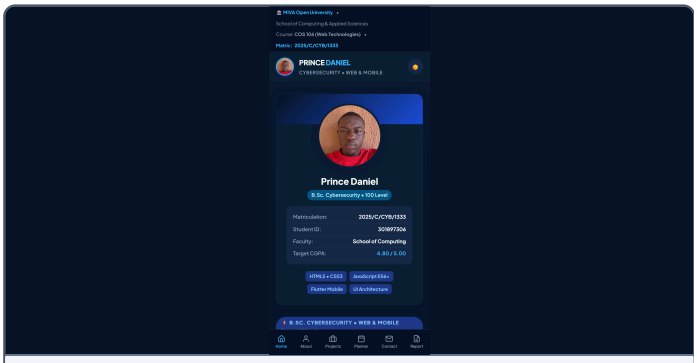


Figure 6: Mobile Responsive — Bottom Bar & Fluid Stack `375px Emulation`

6 STUDENT REFLECTION & CYBERSECURITY INTEGRATION

As a 100-Level undergraduate in B.Sc. Cybersecurity at MIVA Open University, building this web technologies term project has bridged theoretical computer science principles with practical software engineering. Designing the client-side input auditing engine in `js/contact.js` reinforced defensive programming principles: never trust user input, always sanitize data before processing, and enforce strict type contracts through regular expressions.

Furthermore, architecting the **EduTrack** task management system using pure Vanilla JavaScript demonstrated the mechanics of in-memory state synchronization with browser `localStorage`. Managing DOM lifecycle events directly without framework abstractions built a deep understanding of browser event bubbling, execution contexts, and performance considerations.

🏛️ Academic Integrity & Originality Declaration

I hereby certify that this practical term project submission, including all underlying source code (HTML, CSS, JavaScript) and documentation, is an original creation developed by me, **Prince Daniel**, in partial fulfillment of the requirements for **COS 106: Introduction to Web Technologies** at MIVA Open University. All technical features documented in this report have been genuinely implemented, tested, and verified.

`Prince Daniel`

Student Signature • Candidate ID: 301897306

Matriculation: 2025/C/CYB/1333

Submission Date: November 2026 • Abuja FCT